CLOUD PATTERN

<SIBS_M2C_PAT_OPE_DEL_AWS_001_P_ Pipeline CICD STLA_V01.2.docx >

Reference	SIBS_M2C_PAT_OPE_DEL_AWS_001
Published Version	01.2
Apply Date	2022-09-20
Document Classification	Stellantis Internal Usage C1-Internal

Delivery Pattern : Pipeline STLA CI/CD used for SFS AWS

1 Document management Information

Storage source document : **Storage published document (pdf)** :		
Author : (entity) (SFS/SIBS/xxxx**)**	**Reading /Approbation :** **DA Cloud**	**M2C Pattern Manager : to be defined** - **Pilot the review** - **Pilot the modification**

2 Document Management Approbation

who	Date approved	Approval Date and Storage proof
DA CLOUD & M2C pattern Manager		**PDFstored in docinfo repository :** https://docinfogroupe.stellantis.com/ead/dom/1001773080.fd **Source in docinfo repository :** **https://docinfogroupe.stellantis.com/ead/dom/1001765608.fd**

3 Index of révisions

Version (docinfo)	Date of revision	Revised Chapter	Reason	Editors
01.0	2022-07-31	All	Document creation	N .Cernicchiaro
01.1	2022-09-23	All		A. Pierre
01.2	2024-09-01	All	Annual review + correction on “operational diagram CI /CD”	NHA

Table des matières

1 Document management Information 2
2 Document Management Approbation 2
3 Index of révisions 3
1 Introduction and scope (Resume/commentaries) 4
2 Description of the pattern 4
Functional overview pipeline CI/CD : CI/CD (psa-cloud.com) 4
Organisational view and related environments 5
Roles 5
Articulation of processes 5
Role privileges and CIQ environments 6
Focus on Teamcity: 8
Schema operationnel CI /CD : 9
3 Maturity Status of the pattern (to be defined) 10
4 Governance attached to that current document 11
Programmed revision : 11
Modification/enrichment : 11
5 Index, references and glossary 11
5.1 Useful documentation & Guides & references 11
5.2 Glossary 11
5.3 REFLEX application references 12
5.4 Figures table 12

1 Introduction and scope (Resume/commentaries)

- *Context of the pattern*
  - homemade application with or without containerised component and developed according to SFS principles
  - hosted on AWZ SFS LZ

That pattern explains the pipeline STLA CI/CD to be used by an application Team project for deploying

- the application code,
- the application build
  - Especially for containerised applications, the docker image to the STLA Artifactory as a reference and a copy to the applicative project AWS ECR ,
- the application Infra as Code with terraform language .

- *Associated Directive* : NFR-7002

- *Distribution List*

This document is intended for

- **the IT Designer, the IT Integrator, the IT development Team, the OPS teams and DEVOPS teams**
- **the PGAs of the SFS IS**
- and any partner who should work and deploy on SFS AWS LZ cloud.

2 Description of the pattern

Functional overview pipeline CI/CD : CI/CD (psa-cloud.com)

- Continuous integration (CI):

"Process consists of integrating (building) changes of a project (source code mainly), testing these changes and evaluating quality of code. This process can be done repeatedly."

- Continuous delivery (CD):

"Continuous Deployment is a step-up from Continuous Delivery (practice of automating the entire software release process) in which every change in the source code is deployed to production automatically, without explicit approval from a developer."

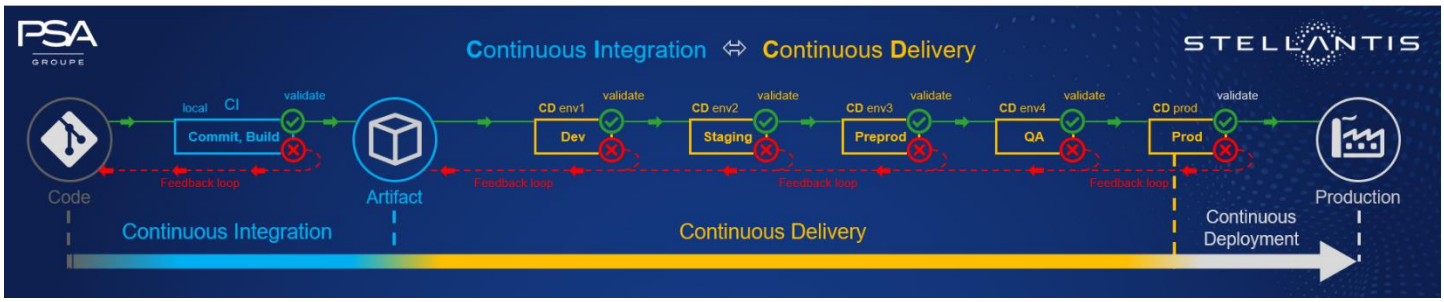


Figure 1 - source STLA CI/CD

Organisational view and related environments

Roles

We have to consider following roles:

- DEV ADMIN: the role of senior developer – in charge of :
  - Creating / maintaining / administrating CIQ DEV environment
  - Administrating GIT organisation for DEV, including creation of repositories
  - Administration of the CI – including setting of build tasks that will permit to publish applicative builds into relevant repository
  - All DEV roles described in the following bullet
- DEV : the role of "simple" developer – in charge of :
  - Update the applicative source code
  - Execute the CI for publishing applicative builds
  - Execute the CD of DEV in order to update the DEV environment.
- OPS : in charge of:
  - Update OPS sources containing "infra as code" language
  - Administration of the non production CD – including setting of build tasks that will permit the CD to be executed
  - Execute the CD of DEV/PreProd/Stage/… in order to update the non production environments.
- RELEASE : in charge of:
  - Administration of the production CD – including setting of build tasks that will permit the CD to be executed
  - Execute the CD of Production in order to update the production environment.

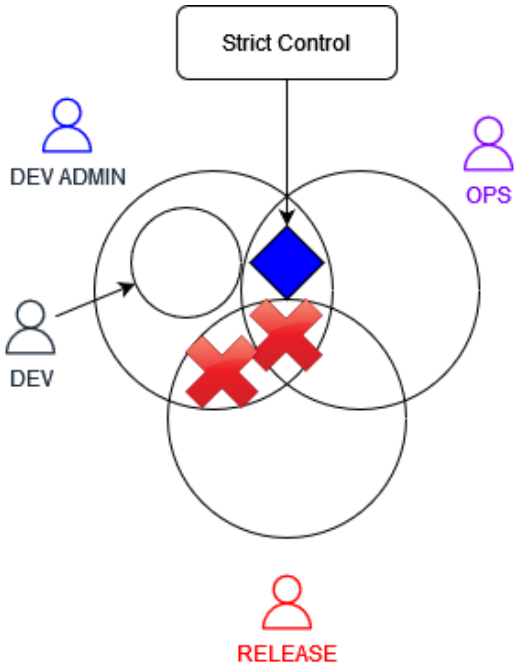


Figure 2 – roles - source: SIBS_M2C_DIR_OPE_DEL_AWS_001 – version 2022-09

Articulation of processes

The CI structure is based on the following:

- Based of source code of the application – hosted through a Git repository in **github-psa-cloud**
- Executed through **Teamcity** – in charge of building the applicative components to be deployed and publish them
- And resulting in a publication of the applicative build into **Artifactory**

Two main processes have to be considered for the CI:

- Publication of a DEV application build: its purpose is to be used on the DEV environment.
- Publication of a RELEASE application build: its purpose is to used on all environment except DEV. The idea is to test the same version of the application on the various non production environment (PreProduction, QA, Stage, …) before to push it into Production.

The CD structure is based on the following:

- Based of source code of the description of infrastructure – defining the **AWS services** used and described with **Terraform** language. The source code is hosted through a Git repository in **github-psa-cloud**
- Taking in input the applicative build hosted in **Artifactory**
- Executed through **Teamcity** – in charge of deploying the infrastructure in AWS and deploying the application in this infrastructure.
- The result is a running application in the expected environment.

The main variation of processes are the following for the CD:

- Non production environments: can be executed by a "DEV Admin" additional to an OPS. The CD of DEV environment itself can be executed by a DEV profile.

- Production environment: CD can be executed by a RELEASE profile only, after control of deliverables having been tested on non-production environments.

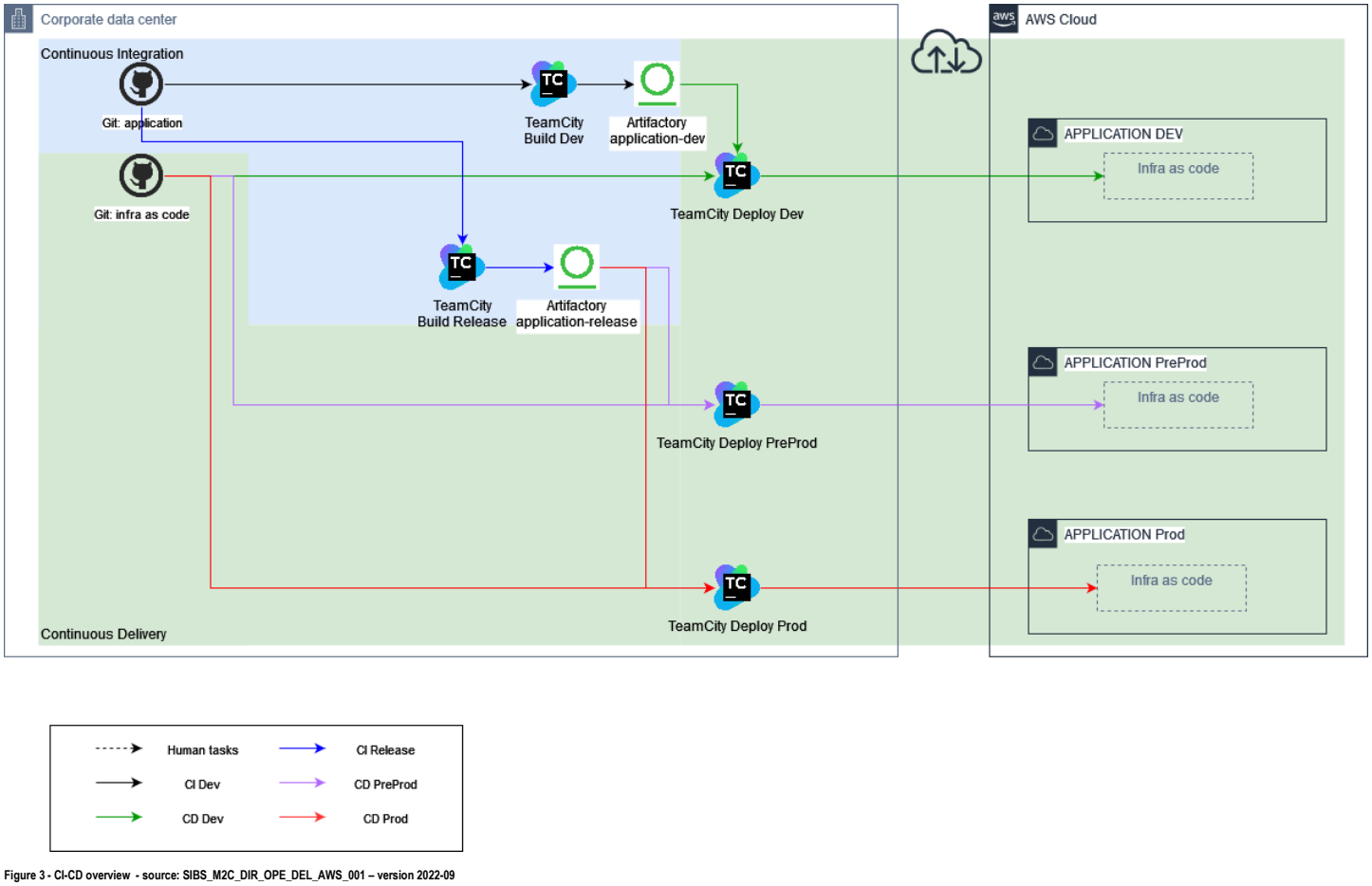


Figure 3 - CI-CD overview - source: SIBS_M2C_DIR_OPE_DEL_AWS_001 – version 2022-09

Role privileges and CIQ environments

The grants associated to each role are defined in the SFS right matrix for CI-CD

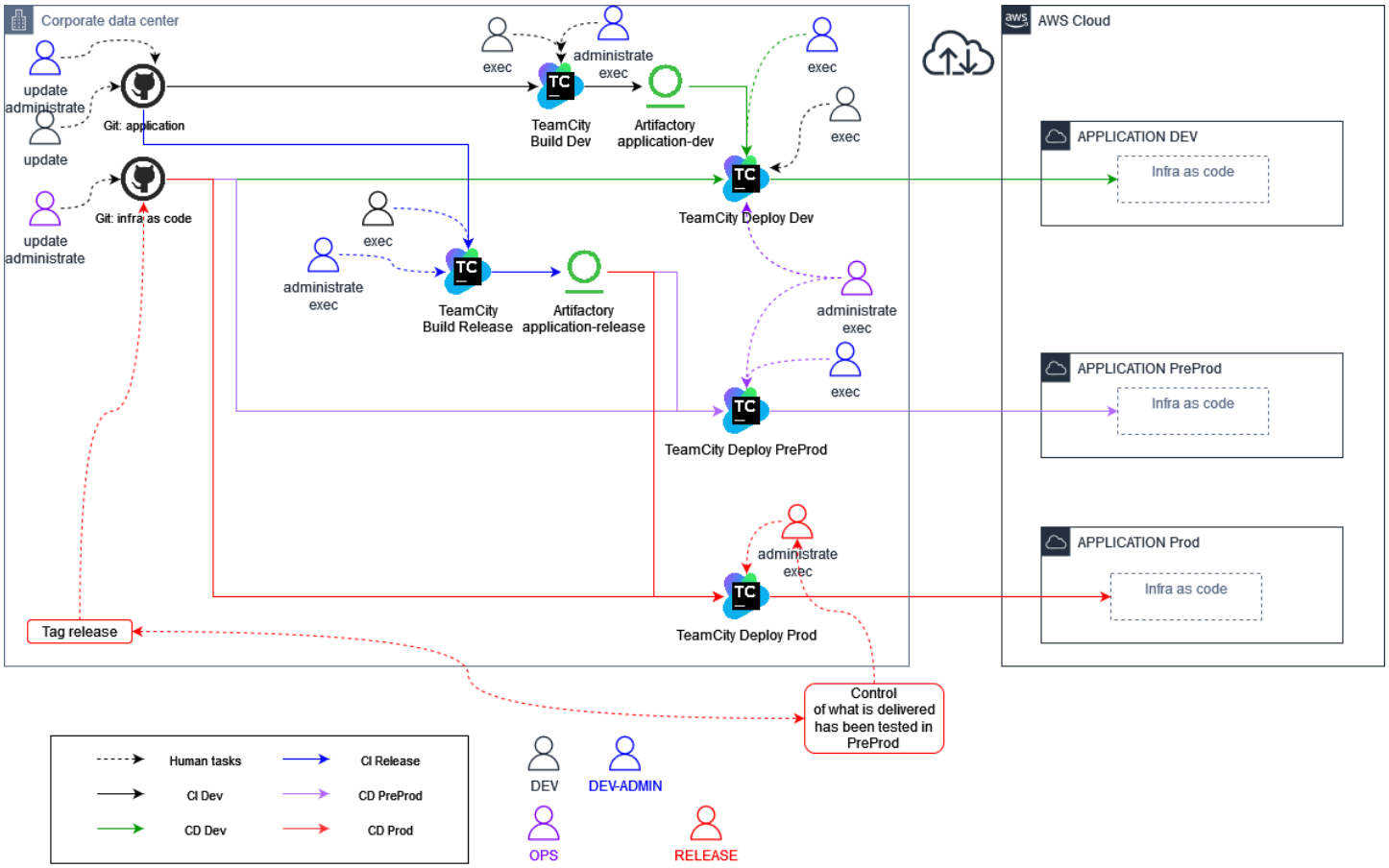


Figure 4 - CI-CD roles - source: SIBS_M2C_DIR_OPE_DEL_AWS_001 – version 2022-09

In order to permit the separation of tasks, one CIQ environment is necessary for each set of rights.

Consequently, we'll have to consider:

- One environment for applicative source
- One environment for infra as code and more generally ops sources
- One environment for the CI
- One environment for the CD for each target environment
  - Note: the non production/non dev environment (ie the preproduction, QA, staging, UAT, etc.) can be on a common CIQ environment depending on the organisation relative to the application.

Here is an example of what may have to be mentioned in the right matrix:

CIQ environments
env <PRD>00** for applicative sources
env <PRD>01** for ops sources
env <PRD>02** for Continuous integration (CI)
env <PRD>03** for DEV deploy
env <PRD>04** for PREPROD deploy *+ UAT/Stage/QA/… if same right policy*
env <PRD>05** for PROD deploy
GitHub access
Access to applicative sources
Access to OPS sources

Focus on Teamcity:

The purpose of Teamcity is to execute automated tasks; examples:

- Build, compilation
- Publication of builds
- Raise to Sonarqube (code quality)
- Execution of unit tests
- Deploy application
- …

We have to distinguish activities related to teamcity:

1) Preparation: the user with a role “admin” on CIQ environment point of view do write the script and configure the tasks relative to its project
2) Launch: the launch of a given task can be done:
    a. Manually from a user with “dev” (on CIQ perspective)
    b. Triggered by an update in source code
    c. Triggered by another teamcity task
3) Execution:
    a. The task will retrieve a source from a VCS
    b. Then the task can do many things: command lines, node/npm/maven executions, docker, SFTP upload/download, aws commands…
    c. Some of the typical purposes are:
        i. Publishing a build into artifactory / ECR
        ii. Reading a build from artifactory
        iii. Deploy infra as code
        iv. Deploy a build on premise
        v. Raise the source to SonarQube for quality analysis
4) After the execution:
    a. The user can read the logs directly into Teamcity
    b. Some execution reports can be notified by e-mail; the notification can be limited to failure
    c. Somme pull request / merge can be raised to git
    d. …

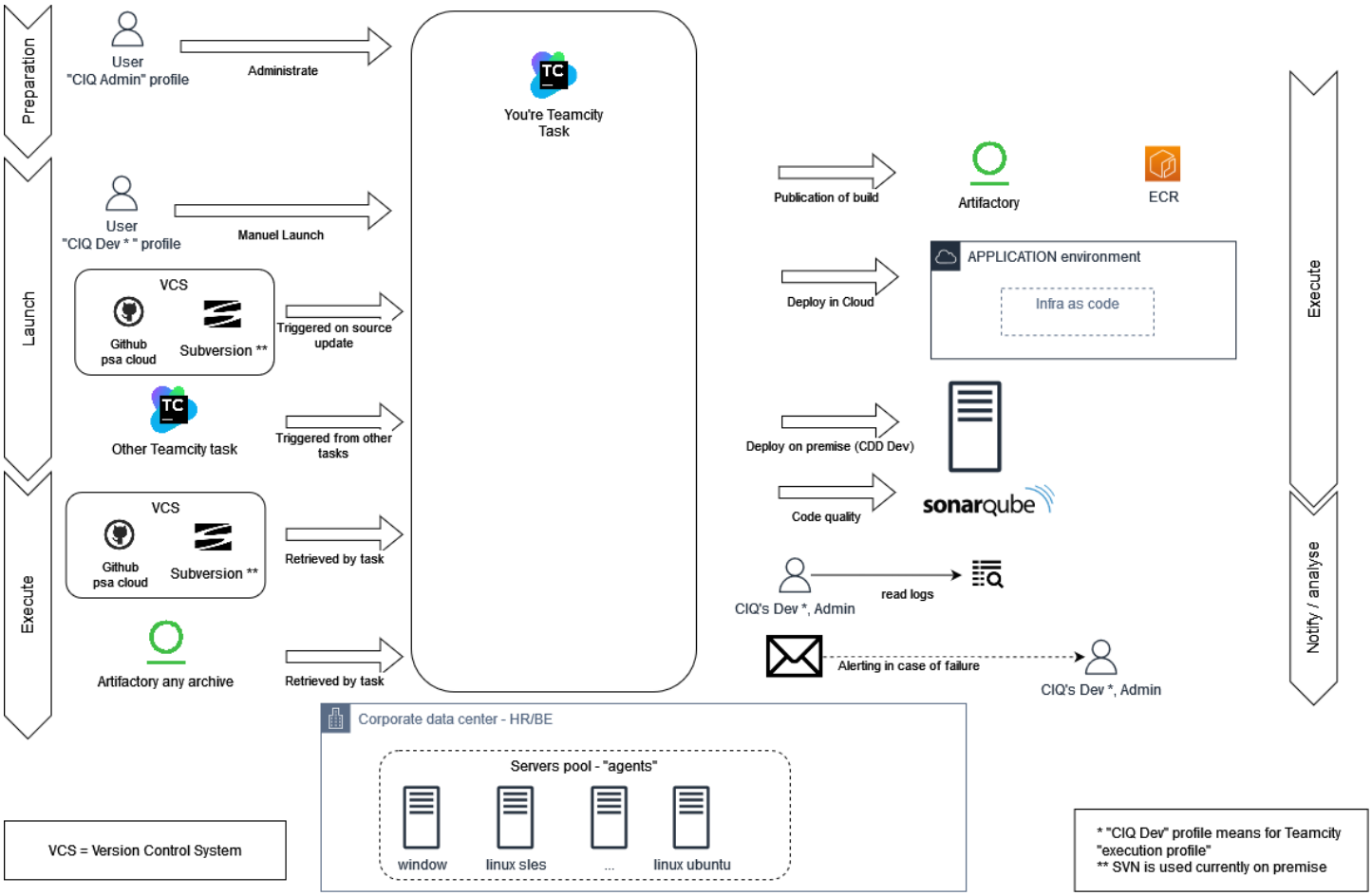
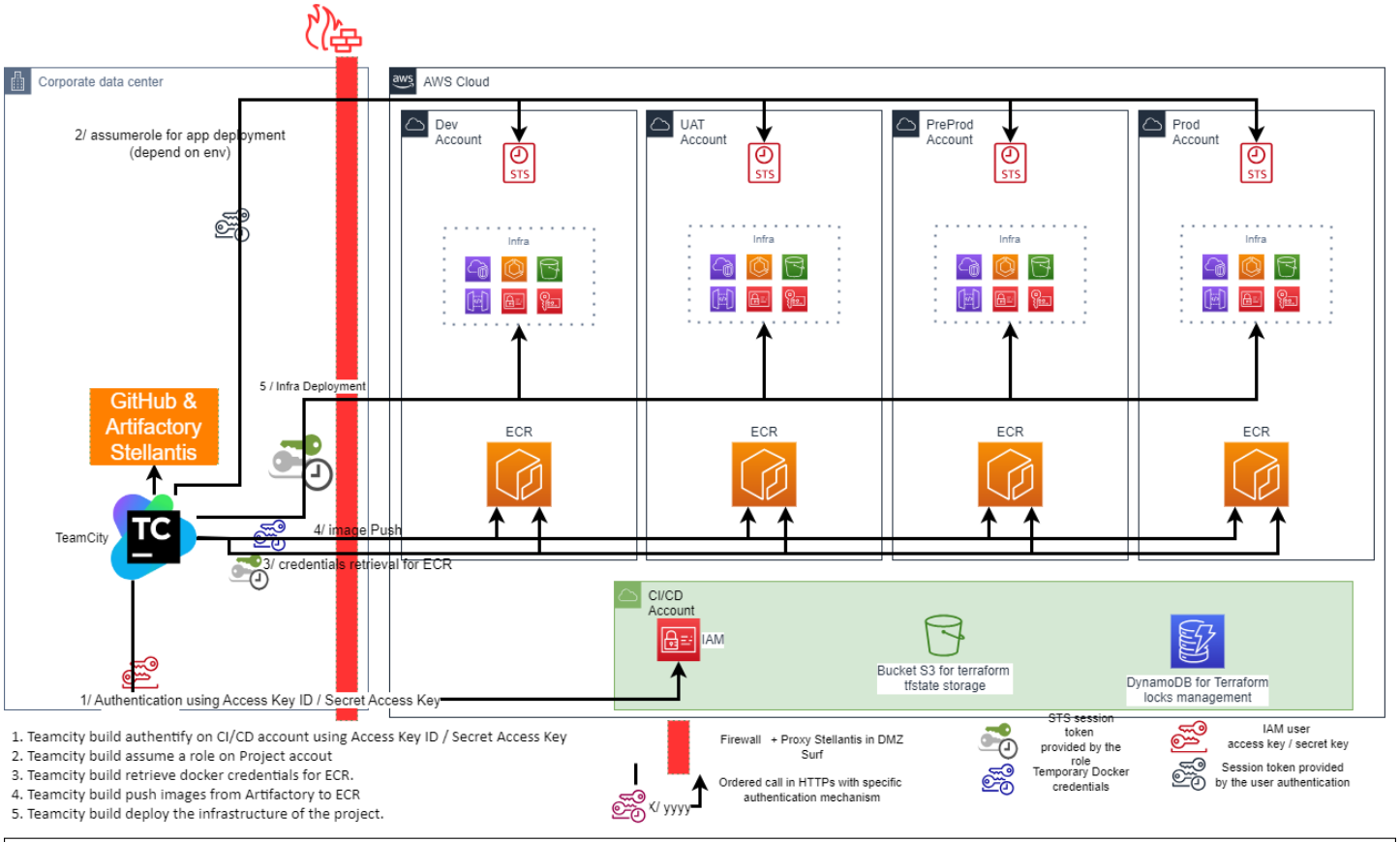


Figure 5 - Teamcity focus - source: SIBS_M2C_DIR_OPE_DEL_AWS_001 – version 2022-09

Operational schema CI /CD :



CICD SFS: AWS deployment authentication 2022/07/28

Figure 6 SIBS_M2C_PAT_OPS_DEL_AWS_001_sche 2_CI/CD Pipeline and access

3 Maturity Status of the pattern (to be defined)

Tested (POC)	**x**					
Testing by Pilot project	**x**					
Production						
Massive deployment						

4 Governance attached to that current document

Programmed revision :

Frequence : one per year and under Pilot document manager and DA Cloud

Modification/enrichment :

Any modification must be addressed to M2C team and the owner of the pattern for studing the demand.

According to the results, the document would be updated, approved and published as a new version (docinfo).

5 Index, references and glossary

5.1 Useful documentation & Guides & references

Cloud Project preparation standard (SIBS)	https://docinfogroupe.stellantis.com/ead/doc/ref.20373_21_07297/v.dp/fiche	M2C Center
SIBS CI/CD cloud Matrix	**Cf doc Aymerick (à renommer)** http://docinfogroupe.inetpsa.com/ead/doc/ref.20373_22_04504/v.vc/fiche http://docinfogroupe.inetpsa.com/ead/doc/ref.20373_22_04503/v.vc/fiche	M2C Center
PSA development Guidelines	**PSA Development Guidelines (psa-cloud.com)**	STLA (collaborative)
TCC dependence (SIBS)	Lien vers le help TCC cloud (SIBS)	M2C center
CI/CD Site (STLA)	**CI/CD (psa-cloud.com)**	STLA
CIQ	http://ciq.inetpsa.com	
Teamcity (in CIQ documentation)	http://ciq-documentation.inetpsa.com/quickstart/teamcity.html	

5.2 Glossary

CI	Continuous Integration
CD	Continuous deployment
Docinfos	Stellantis IS for document management (publication , exposure)

Teams	**Workspace for collaborative work. (draft document)** SFS Move to cloud Teams **SFS architecture technique** Etc…

5.3 REFLEX application references

REFLEX name	endpoint
CIQ	http://ciq.inetpsa.com
GITHUB	https://github.psa-cloud.com/
ARTIFACTORY	http://repository.inetpsa.com/ui/

5.4 Figures table

Figure 1 - source STLA CI/CD 4
Figure 2 – roles - source: SIBS_M2C_DIR_OPE_DEL_AWS_001 – version 2022-09 5
Figure 3 - CI-CD overview - source: SIBS_M2C_DIR_OPE_DEL_AWS_001 – version 2022-09 6
Figure 4 - CI-CD roles - source: SIBS_M2C_DIR_OPE_DEL_AWS_001 – version 2022-09 7
Figure 5 - Teamcity focus - source: SIBS_M2C_DIR_OPE_DEL_AWS_001 – version 2022-09 9
Figure 6 SIBS_M2C_PAT_OPS_DEL_AWS_001_sche 2_CI/CD Pipeline and access 10